



Emergent multiferroic altermagnets and spin control via noncollinear molecular polarization

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Altermagnets, with spin splitting and vanishing magnetization, have been attributed to many fascinating phenomena and potential applications. In particular, integrating ferroelectricity with altermagnetism to enable magnetoelectric coupling and electric control of spin has drawn significant attention. However, its experimental realization and precise spin manipulation remain elusive. Here, by focusing on molecular ferroelectrics, the first discovered ferroelectrics renowned for their highly controllable molecular polarizations and structural flexibility, we reveal that these obstacles can be removed by an emergent multiferroic altermagnet with tunable spin polarization in a large class of fabricated organic materials. Using a symmetry-based design and a tight-binding model, we uncover the underlying mechanism of such molecular ferroelectric altermagnets and demonstrate how noncollinear molecular polarization can switch the spin polarization on and off and even reverse its sign, as detectable by the magneto-optical Kerr effect. From the first-principles calculations, we verify the feasibility of these materials in a series of well-established hybrid organic-inorganic perovskites and metal-organic frameworks. Our findings bridge molecular ferroelectrics and altermagnetic spintronics, highlighting an unexplored potential of multifunctional organic multiferroics.

altermagnets, multiferroics, magnetoelectric coupling, spintronics, ferroelectrics, molecular polarization

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1 Introduction

Electrical control of magnetism is a fundamental challenge in condensed matter physics, with significant implications for high-density data storage and energy-efficient spintronic

devices [1-3]. Magnetoelectric coupling, which connects magnetic and electric ordering, is crucial in multiferroics [4-7]. Traditionally, multiferroics leverage antiferromagnets due to their insulating nature, absence of stray fields, and ultrafast dynamics [8-11]. However, the lack of spin polarization in antiferromagnets limits their potential in spintronics. The growing interest in altermagnets (AM) [12-34], which unify the spin polarization of ferromagnets (FM) and the zero

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net magnetization of antiferromagnets (AFM), significantly broadens the avenues for realizing magnetoelectric coupling in multifunctional materials.

Integrating (anti)ferroelectricity with altermagnetism offers a new route toward multiferroics. Recent discoveries about altermagnetoelectric coupling [35–40], enable electric control of magnets, highlighting the potential of AM for developing electrically controlled spintronic and multiferroic devices. The key to such integration is the ability of (anti)ferroelectricity to connect opposite-spin sublattices via rotation, R , symmetry rather than translation, t , or inversion, I , symmetries. Several strategies, including antiferroelectricity [35], lattice distortion [36–38], and layer sliding/stacking [39, 41–43] have been explored to realize this symmetry-driven coupling, yet experimental realization and precise spin control remain elusive.

Molecular ferroelectricity, as the first ferroelectricity discovered in 1921 [44], presents a versatile alternative [45] for realizing multiferroic altermagnets. Unlike conventional ferroelectrics, molecular ferroelectrics derive their polarization from the alignment of molecular dipoles, providing exceptional structural tunability, mechanical flexibility, and multifunctionality [46]. The ability to precisely engineer molecular size, shape, charge, and polarity enables versatile control over molecular polarization, P_M [46, 47], which has been experimentally realized using external stimuli such as electric field [46–53], pressure [49, 50], temperature [49, 50], and light [54]. Moreover, the weak spin-orbit coupling (SOC) and hyperfine interactions, intrinsic to light elements such as C, H, O, and N, contribute to excellent spin properties, including long spin-diffusion lengths and relaxation times [45, 49, 55, 56]. Despite these advantages, integrating AM with molecular ferroelectricity remains unexplored, representing a significant gap in the advancement of molecular multiferroics and their devices.

In this work, we present a universal design framework for molecular ferroelectric altermagnets (MFEAM), which combines the momentum-dependent spin splitting of AM with the structural flexibility of molecular ferroelectricity. Using spin-group theory [57–62], we identify the critical role of noncollinear P_M in creating R symmetry, connecting spin sublattices without t or I symmetries. Molecular systems inherently support this symmetry through their controllable polarizability and spatial flexibility. Employing a tight-binding (TB) model, we show that the P_M distribution governs the magnetic properties, and that twisting P_M can effectively toggle spin polarization, P_S , on/off and even reverse its sign. Guided by this theoretical framework, we use first-principles calculations to propose a series of promising MFEAM candidates, ranging from hybrid organic-inorganic perovskites (HOIPs) to metal-organic frameworks (MOFs). These find-

ings lay the groundwork for advancing molecular multiferroics and developing the next-generation multifunctional devices.

2 Results

2.1 Design principle

Our proposed framework for MFEAM is illustrated in Figure 1, where the basic unit consists of magnetic atoms forming AFM order and their surrounding molecules with P_M . When P_M are collinear, either parallel (PP) or antiparallel (AP), the resulting state is a conventional AFM characterized by vanishing P_S and spin-group symmetries $[C_2||t]$ or $[C_2||I]$ [12, 19] as shown in (Figure 1(b)). Here, C_2 denotes the two-fold rotation operator in spin space, while t and I represent translation and inversion operations in crystal structures, respectively. In contrast, when P_M become non-collinear (NP), an AM state emerges, featuring finite P_S and symmetry characterized by $[C_2||R]$ (Figure 1(c)). Notably, reversing a single P_M flips the P_S sign. Thus, by simply twisting P_M in MFEAM, one can tune its magnetism, control the presence of P_S , and reverse its sign, all without changing the magnetic order. Since the P_M can be twisted by electric field [46–53], temperature [49, 50], and light [54], this tunable behavior enables promising spintronic and multiferroic functionalities (Figure 1(d)).

2.2 Effective model

To elucidate the mechanism of our proposed MFEAM, we construct an effective tight-binding (TB) model on a general 2D square lattice with nested AFM order, as illustrated in Figure 2. This setup incorporates the essential components of our proposal: AFM sublattices with collinear (noncollinear) P_M connected by t/I (R) symmetry. Our symmetry analysis has revealed that the emergence of AM (AFM) depends on the inequivalence (equivalence) of the magnetic sublattices governed by the P_M distribution. To accurately capture such P_M influence, the model necessitates hopping terms up to the third-nearest neighbor. Thus, the minimal TB model for MFEAM is

$$H = \left(\sum_{i,j} (f_i^{\eta_j} c_i^\dagger c_{i+\eta_j} + g_i^{\kappa_j} c_i^\dagger c_{i+\kappa_j} + h_i^{\delta_j} c_i^\dagger c_{i+\delta_j}) + \text{H.C.} \right) + M_{A,B} \sum_{i \in A,B} c_i^\dagger \sigma_z c_i. \quad (1)$$

Here, $c_i^\dagger$ and c_i represent electron creation and annihilation operators at site i . The parameters $f_i^{\eta_j}$, $g_i^{\kappa_j}$, and $h_i^{\delta_j}$ describe

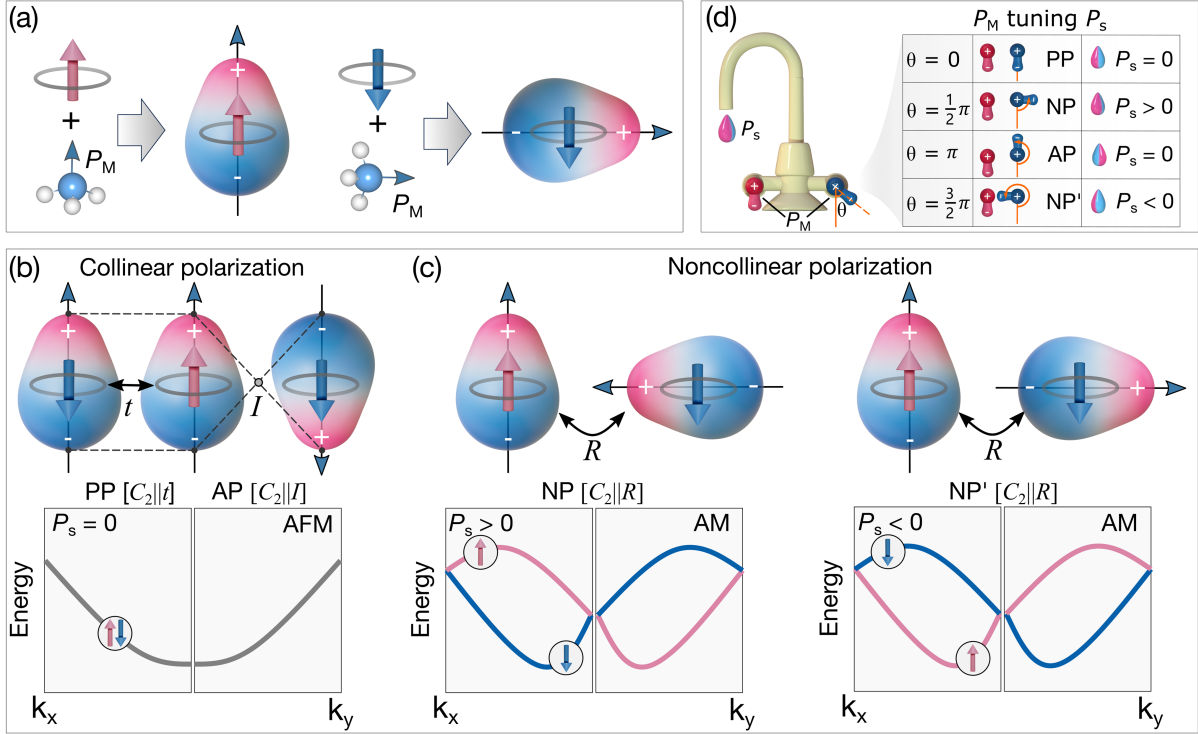


Figure 1 (Color online) Design principle of MFEAM with spin control via P_M twist. (a) The fundamental building block consists of magnetic atoms (arrows with circles) and polar molecules with P_M , directed from the negative to the positive charge center. (b) Parallel (PP) and antiparallel (AP) P_M establish magnetic sublattices related by t and I symmetry, resulting in conventional AFM. (c) Noncollinear (NP) P_M configurations connect magnetic sublattices via R symmetry, giving rise to AM with finite P_S (assumed $P_S > 0$ for clarity). Reversing one P_M in the NP configuration (denoted as NP') flips P_S . (d) Schematic illustration of tuning P_S through P_M , where P_M is indicated by the faucet handle, and P_S by water drops with red (blue) for spin-up (-down). Twisting P_M enables the switching P_S on/off and reversing its sign.

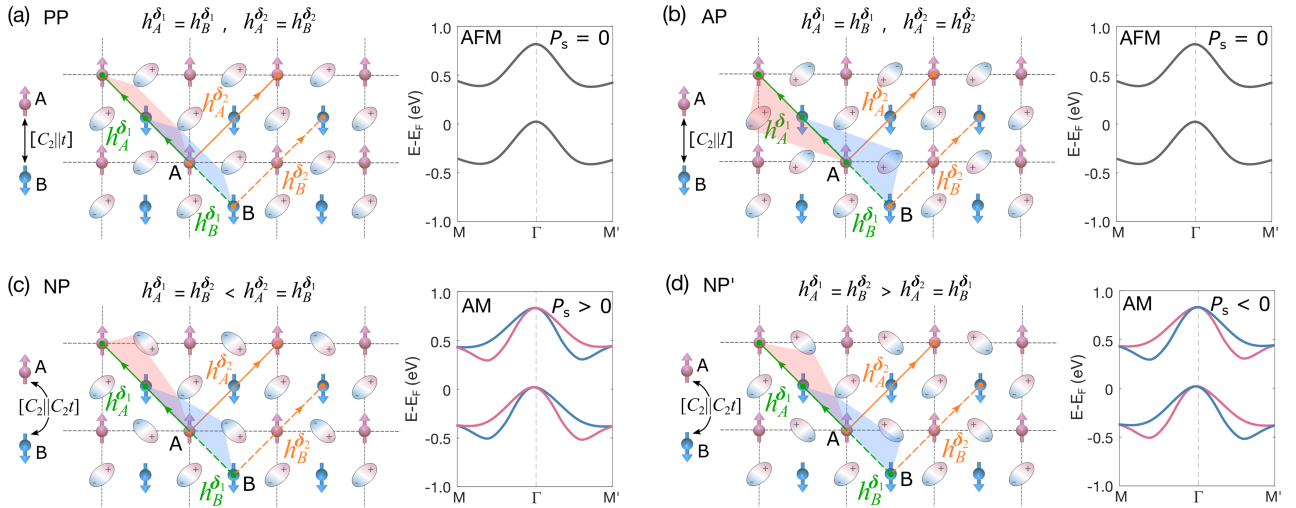


Figure 2 (Color online) Effective model. Schematic of 2D square magnetic sublattices A and B surrounded by polar molecules (ellipses), forming PP, AP, NP, and NP' configurations with spin lattices connected by symmetry t , I , C_2t , respectively. The solid (dashed) arrows indicate 3NN hopping vectors ($\delta_{1,2}$) in A (B) sublattice, where the red (blue) shades illustrate how its hopping strengths $h_A^{\delta_{1,2}}$ ($h_B^{\delta_{1,2}}$) are affected by P_M distribution. The other hopping vectors $\delta_{3,4}$ are not shown as their hopping strengths along opposite directions are identical ($h_{A,B}^{\delta_{3,4}} = h_{A,B}^{\delta_{1,2}}$). Spin-resolved bands are displayed in black (spin-degenerate), red (spin-up), and blue (spin-down), calculated using the TB model. The critical 3NN parameters are (a) and (b) $h_A^{\delta_{1,3}} = 0.03$ eV, $h_{A,B}^{\delta_{2,4}} = 0.08$ eV; (c) $h_A^{\delta_{1,3}} = h_B^{\delta_{2,4}} = 0.03$ eV, $h_A^{\delta_{2,4}} = h_B^{\delta_{1,3}} = 0.08$ eV; (d) $h_A^{\delta_{1,3}} = h_B^{\delta_{2,4}} = 0.08$ eV, $h_A^{\delta_{2,4}} = h_B^{\delta_{1,3}} = 0.03$ eV, with $M_A = -M_B = 0.4$ eV. All other parameters and model details including high-symmetry k -points are provided in Supplementary Note S1.

the hopping between site i and its first (NN), second (2NN), and third (3NN) nearest-neighbors, which are connected by the vectors of η_j , κ_j , and δ_j , respectively. $M_{A,B}$ denotes the AFM on-site exchange field on sublattices A and B .

According to our TB model analysis, neither NN nor 2NN hoppings contribute to AM as they fail to induce the necessary sublattice inequivalence regardless of the P_M distribution, as discussed in the Supplementary Note S1. In contrast, 3NN hoppings between sublattices can be either equivalent or inequivalent depending on the P_M configuration. Specifically, in the PP and AP configurations, the system retains the symmetries $[C_2||t]$ and $[C_2||I]$, ensuring sublattice-equivalent 3NN hoppings ($h_A^{\delta_{1-4}} = h_B^{\delta_{1-4}}$). This leads to conventional AFM, as confirmed by the spin-degenerate bands in Figure 2(a) and (b). However, the situation changes completely in the NP configuration. The noncollinear P_M breaks both t and I symmetries, leaving the sublattices connected by a rotation symmetry, C_2t . This leads to inequivalent 3NN hoppings, with $h_A^{\delta_{1,3}} = h_B^{\delta_{2,4}} < h_A^{\delta_{2,4}} = h_B^{\delta_{1,3}}$, and gives rise to the desired MFEAM, as evidenced by spin-polarized bands (Figure 2(c)). Remarkably, reversing a single P_M to form the NP' configuration interchanges the 3NN hopping parameters ($h_A^{\delta_{1,3}} = h_B^{\delta_{2,4}} > h_A^{\delta_{2,4}} = h_B^{\delta_{1,3}}$), thereby producing an AM state with reversed P_S compared with that in the NP configuration (Figure 2(d)).

These TB model results clearly reveal the underlying mechanism of our MFEAM design, demonstrating how P_M configurations govern magnetism and spin polarization. Since all above four configurations can be realized simply by twisting P_M , this provides a compelling approach for toggling P_S on and off or reversing its sign without changing the magnetic order. This capability holds great promise for spintronic applications and nonvolatile multiferroic devices [1-7].

2.3 Material realization

Our design principle offers a universal framework for realizing MFEAM based on symmetry analysis, which has been further extended into a Hamiltonian description via our effective model. To bridge theory and experiment, the key criteria for potential candidates are the presence of AFM order and noncollinear P_M distributions. A promising strategy is to start with AFM materials and incorporate polar molecules to induce the necessary P_M configurations, which can be selectively controlled during fabrication and tuned through external stimuli such as electric fields, pressure, and light [46-54].

Here, we focus on two well-established families of magnetic organic materials: HOIPs and MOFs. These materials are widely studied in energy applications, catalysis, spintronics, and multiferroics, owing to their facile synthesis, structural diversity, and cost-effectiveness [49, 63, 64]. We first

examine magnetic HOIPs with the general formula A_2MX_4 , where A represents an organic molecular cation, M is a divalent transition metal ion that governs the magnetic properties, and X is a halogen anion. The magnetism in HOIPs is primarily dominated by M , for instance, Mn commonly induces AFM order [65], while its P_M distribution mainly depends on A [48]. Notably, all our proposed P_M configurations, PP, AP, and NP/NP' with AFM order have been achieved experimentally through appropriate selection of A , M , and X , combined with controlled synthesis conditions [46-50].

As a representative example, we consider layered van der Waals material $[MA]_2MnCl_4$ (where $MA = CH_3NH_3$, methylammonium) due to its relatively simple crystal structure (Figure 3(a)) and its experimentally observed AFM ground state [66, 67]. Our first-principles calculations focus on the "organic-inorganic-organic" 2D monolayer, each monolayer consists of corner-sharing $[MnCl_6]$ octahedra surrounded by MA molecules, which dictate the properties of $[MA]_2MnCl_4$. The P_M distribution is controlled by the orientation of the MA molecules. Here, we constructed four different configurations corresponding to our design principle, as shown in Figure 3(b)-(e). A more detailed discussion of additional configurations can be found in Supplementary Note S2. Experimentally, HOIP materials with various molecular orientations have already been fabricated and successfully controlled using external electric fields, pressure, and light [46-54], providing strong experimental support for the realization and control of MFEAM.

PP configuration belongs to the $Pmm2$ space group, where the presence of t symmetry restores conventional AFM (Figure 3(b)). Similarly, in the AP configuration, the system adopts the $Cmmm$ space group, where its I symmetry enforces AFM (Figure 3(c)). These symmetry-dependent magnets can be further understood through the P_M -induced distortions. The presence of P_M affects the tilt of $[MnCl_6]$ octahedra, modifying the charge distribution and altering symmetry between spin sublattices, ultimately determining the P_S . Conversely, when the MA molecules align into a NP configuration, the system transitions to the $Pmc2_1$ space group, where a $C_{2x}t$ symmetry connects the magnetic sublattices, inducing AM, as evidenced by the calculated band structure showing a finite P_S (Figure 3(d)). Furthermore, flipping a single MA molecule to form the NP' configuration reverses the direction of P_S (Figure 3(e)), fully consistent with our TB model. The detailed symmetry analysis of different configurations is provided in the Supplementary Note S3, confirming that the variation in symmetry operations induced by the P_M distribution plays a decisive role in the emergence of AM.

Since $[MA]_2MnCl_4$ is composed primarily of light elements, spin-orbit coupling is weak, as confirmed by our calculations in Supplementary Note S4, facilitating experi-

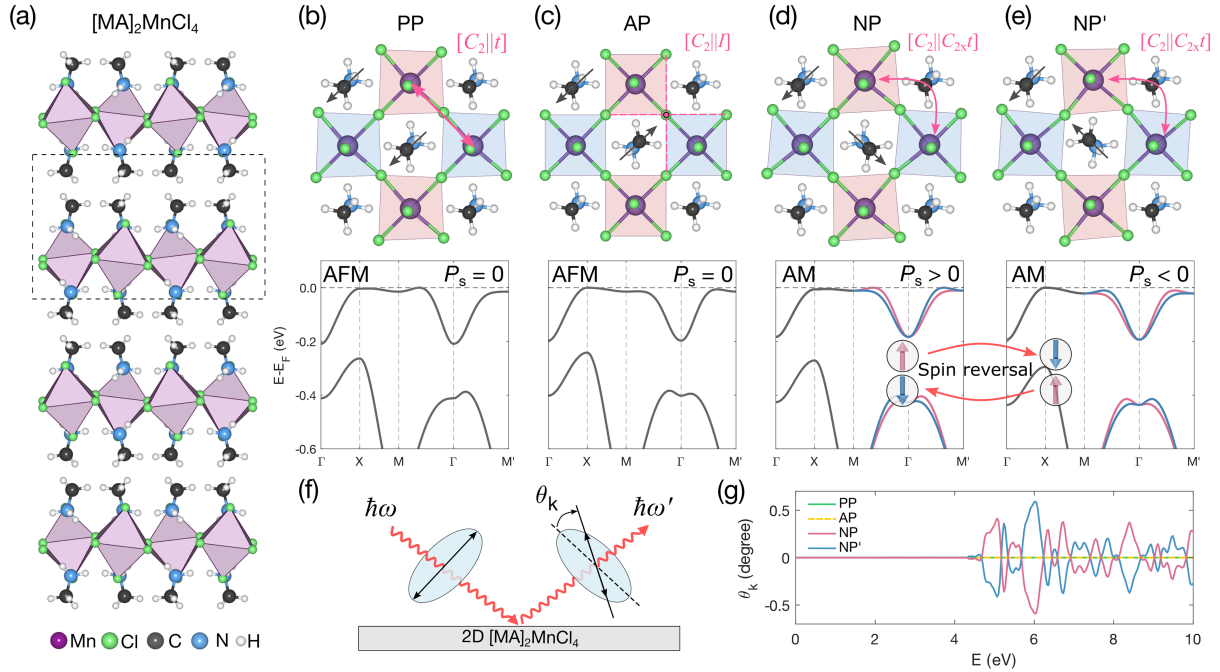


Figure 3 (Color online) MFEAM realization in $[\text{MA}]_2\text{MnCl}_4$. (a) Crystal structure of layered $[\text{MA}]_2\text{MnCl}_4$, with the monolayer unit highlighted by a dashed box. (b)–(e) Top views of the $[\text{MA}]_2\text{MnCl}_4$ monolayer in the PP, AP, NP, and NP' configurations, along with the corresponding calculated spin-resolved electronic band structures. (f) Schematic illustration of the magneto-optical Kerr effect measurement setup. (g) Calculated MOKE signals for each configuration, characterized by the Kerr rotation angle θ_k .

mental identification of the altermagnetic spin splitting via ARPES, nanoscale imaging, or spin transport [25–27]. Here, we propose probing the characteristics of altermagnetism through the detection of the magneto-optical Kerr effect (MOKE) [68–71]. As shown in Figure 3(g), the PP and AP configurations exhibit negligible Kerr signals. In contrast, a pronounced Kerr response is observed in the NP/NP' configurations, which can be attributed to the anisotropic optical conductivity induced by the breaking of time-reversal symmetry. It is noteworthy that the Kerr angle magnitude serves as an indicator of spin splitting strength, as reported in previous studies [72, 73]. Furthermore, the Kerr angle exhibits a sign reversal between the NP and NP' configurations, consistent with the reversal of spin splitting observed in the band structures. The demonstrated P_M control of the Kerr response further highlights the robustness of magnetoelectric coupling in the proposed MFEAM.

Our MFEAM design can be extended by substituting MA in $[\text{MA}]_2\text{MnCl}_4$ with other common organic cations such as ethylammonium (EA), dimethylammonium (DMA), and phenylammonium (PA), as illustrated in Figure 4(a). These molecules preserve the Mn-based AFM sublattice and tend to adopt NP configurations, making them promising MFEAM candidates, as confirmed by our calculations (Figure 4(b) and Supplementary Note S5). Notably, replacing MA with PMA in $[\text{PMA}]_2\text{MnCl}_4$ enhances the spin splitting from 26 meV to 41 meV due to the larger molecular size, which induces

stronger distortions in the $[\text{MnCl}_6]$ octahedra. This effect provides a practical means to enhance P_S , making altermagnetic behavior more prominent.

Additionally, replacing the Cl component in A_2MnCl_4 HOIPs with organic anions including $[\text{GUA}]\text{Cu}(\text{HCOO})_3$, $[\text{DMA}]\text{Cu}(\text{HCOO})_3$, $[\text{NH}_4]\text{Cu}(\text{HCOO})_3$ leads to the formation of MOFs, offering another pathway to expand MFEAM candidates as verified in our calculations (Figure 4(c)–(e)). Such large organic anions can introduce stronger structural distortions, further enhancing the altermagnetic spin splitting, which reaches 170 meV in $[\text{DMA}]\text{Cu}(\text{HCOO})_3$, strong enough to support spintronic applications. Although this work focuses on organic materials, our MFEAM design principle can be extended to inorganic systems. For instance, we find the reported multiferroic BaFe_2Se_3 [74] and Pb_2MnWO_6 [75] are inorganic MFEAM, as discussed in Supplementary Note S5.

3 Discussion and conclusion

Our proposed MFEAM bridges two previously separate concepts, molecular ferroelectricity and altermagnetism, into a single material platform, establishing a universal framework for expanding organic multiferroics and realizing novel magnetoelectric coupling. Many of our proposed MFEAM candidates are composed primarily of light elements, making

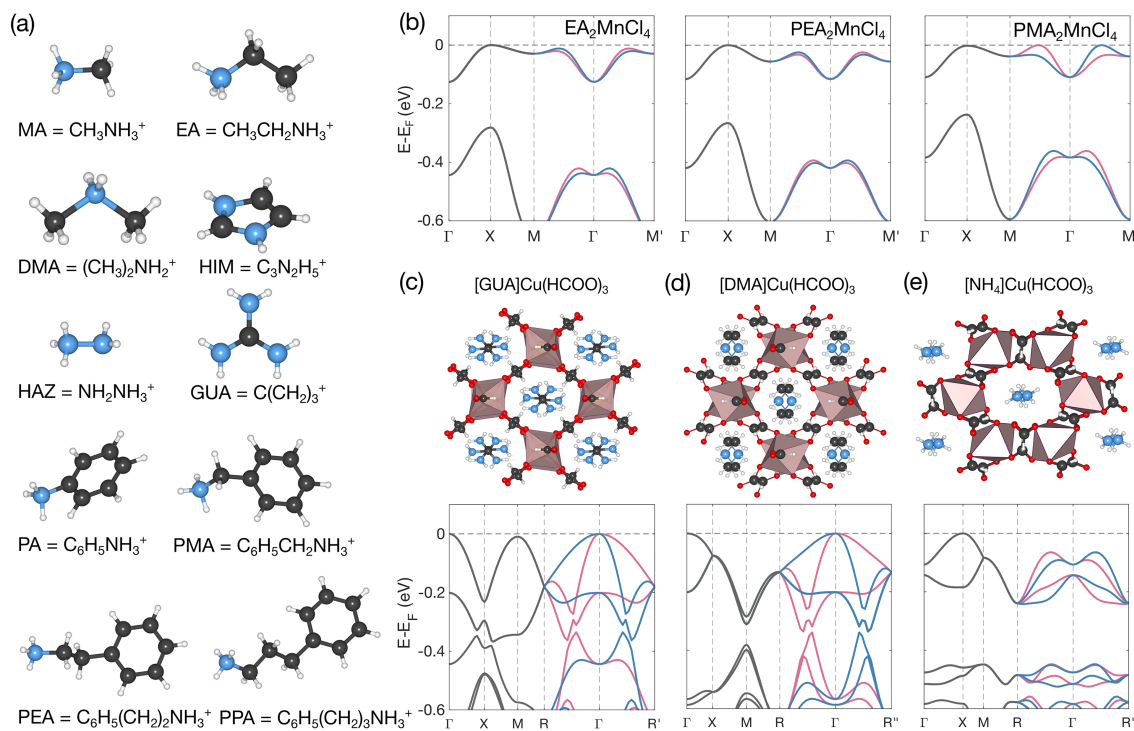


Figure 4 (Color online) MFEAM candidates in HOIPs and MOFs. (a) Structural diversity of organic polar molecules. (b) Calculated band structures of Mn-based HOIP for MFEAM, including $[\text{EA}]_2\text{MnCl}_4$, $[\text{PEA}]_2\text{MnCl}_4$ and $[\text{PMA}]_2\text{MnCl}_4$. (c)-(e) Crystal structures and corresponding bands of MOFs for MFEAM, including $[\text{GUA}]\text{Cu}(\text{HCOO})_3$, $[\text{DMA}]\text{Cu}(\text{HCOO})_3$, $[\text{NH}_4]\text{Cu}(\text{HCOO})_3$.

them ideal for studying organic spin dynamics dominated by nonrelativistic spin splitting rather than SOC effects [76, 77]. Moreover, the P_S in MFEAM can be flexibly controlled by twisting P_M via electric fields, pressure, or light, enabling efficient, zero-magnetic-field spin manipulation. This versatile control paves the way for developing electrically or optically driven spintronic devices. In addition, several of our proposed MFEAM materials are already recognized for their advanced performance in solar energy and catalysis [78-80]. The incorporation of these newly identified multiferroic properties further broadens their scope, positioning them as highly promising candidates for multifunctional applications [81-83].

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Conflict of interest The authors declare that they have no conflict of interest.

Supporting Information

The supporting information is available online at <http://phys.scichina.com> and <https://link.springer.com>. The supporting materials are published as

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